

Total No. of Questions : 4]

SEAT No. :

PF227

[Total No. of Pages : 2

Apr-26/SE/Insem-281

S.E. (AIML) (Insem)

OPERATING SYSTEMS

(2019 Pattern) (Semester-IV) (Theory) (218552)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, or Q.3 or Q.4.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) What is an operating system? Discuss developments leading to Modern Operating Systems. **[5]**

b) Explain the significance of the following shell commands: echo, cut, touch. **[5]**

c) Describe the evolution of the Operating System. **[5]**

OR

Q2) a) Describe with the help of a neat diagram the concept of Time-shared Systems. **[5]**

b) Write a shell script that calculates the sum of integers from 1 to N. **[5]**

c) What is a virtual machine? Explain the concept of virtualization. **[5]**

P.T.O.

Q3) a) With the help of a neat, explain in detail process state transition diagram with 5 states. [7]

b) Consider the Set of Processes with Arrival Time, Burst Time & Priority.[8]

Process	Arrival Time	Burst Time
P1	2	2
P2	0	7
P3	6	3
P4	9	8

Find Average Turnaround Time & Average Waiting Time for SJF (Preemptive) & Round Robin (Time Quantum = 2) scheduling algorithms with the help of Gantt chart

OR

Q4) a) Discuss Process Creation. How new task assigned is the child process? Determine the number of child processes if the parent process executes 4 consecutive forks () commands [7]

b) Discuss with the help of neat diagrams different thread models. [8]

